

Lab 01

Resource Group and Virtual Machine

Your first resource in Azure - created two ways, verified, and deleted

Lab number	01 of the AZ-104 practical series
Maps to	Module 2 (Azure basics) and Module 3 (Virtual Machines)
Exam domain	Deploy and manage Azure compute resources
Duration	35 - 45 minutes
Cost	Free tier eligible. Charges apply only if you skip the cleanup step.
Prerequisites	An active Azure subscription and a browser. Nothing installed locally.

Objectives

- Sign in to the Azure portal and identify your subscription.
- Understand where a resource sits in the Azure hierarchy.
- Create a resource group through the portal.
- Create a Linux virtual machine and connect to it.
- Perform the identical task using Azure CLI in Cloud Shell.
- Verify the deployment and read the cost.
- Delete everything cleanly so that nothing continues to bill.

Rule for every lab in this series

Whatever you create, you delete before you leave. A virtual machine left running overnight will consume your free credit. The final step of every lab is cleanup, and it is not optional.

Before You Click Anything

A resource in Azure never exists on its own. It always sits inside a chain of containers, and understanding that chain now will save you considerable confusion later in the course.

Level	What it is	Key point
Microsoft Entra ID Tenant	Your organisation's identity directory. Holds users and groups.	Identity lives here, not inside the subscription.
Management Group	An optional container used to apply policy across many subscriptions.	Only relevant in larger organisations.
Subscription	The billing boundary. Every resource is billed to exactly one subscription.	If you have no subscription, you cannot create anything.
Resource Group	A logical folder holding related resources that share a lifecycle.	Deleting a resource group deletes everything inside it. Permanently.
Resource	The actual thing - a virtual machine, a disk, a network, a storage account.	Every resource must belong to exactly one resource group.

Coming from AWS?

AWS has no direct equivalent of a resource group. In AWS, resources sit directly under an account. In Azure, the resource group is a real container with real consequences - permissions can be assigned to it, and deleting it removes everything inside. Treat it as a project folder, not a label.

What you will build in this lab

A single Linux virtual machine inside a new resource group. Azure will automatically create the supporting resources around it - a virtual network, a subnet, a network interface, a public IP address, a network security group and a managed disk. Watching Azure create those seven objects from one request is itself a useful part of this lab.

Resource created	Purpose	Created by you or automatically?
Resource Group	Container for everything below	By you, in Step 3
Virtual Machine	The server itself	By you, in Step 4
Managed Disk	The operating system disk	Automatically
Virtual Network + Subnet	The private network the VM sits in	Automatically
Network Interface	Connects the VM to the network	Automatically
Public IP Address	Lets you reach the VM from the internet	Automatically
Network Security Group	Firewall rules controlling inbound traffic	Automatically

Naming convention used throughout this series

Use exactly these names. Consistent naming makes the screenshots in this guide match what you see on your own screen, and matching names make troubleshooting far easier in a room of sixty students.

Item	Value to use
Resource group name	rg-lab01-<your roll number>
Virtual machine name	vm-lab01
Region	Central India
Image	Ubuntu Server 22.04 LTS
Size	Standard B1s (free tier eligible)
Authentication	SSH public key
Username	azureuser

Part A - Using the Azure Portal

The portal is the graphical interface. It is the easiest place to start because every option is visible and labelled. Later in this lab you will do the same thing in three commands.

1 Sign in to the Azure portal

Open a browser and go to the Azure portal address. Sign in with the account attached to your subscription. If you are asked to choose a directory, select the one that holds your subscription.

SHOT 1.1

PASTE SCREENSHOT HERE

Azure portal home page immediately after sign-in, showing the top search bar and the Azure services row.

Figure 1.1 - The Azure portal home page.

What to check before moving on

The top right corner should show your account name and the directory you are signed in to. If it shows the wrong directory, switch it before continuing, or you will create resources in the wrong place.

2 Confirm you have an active subscription

Type **Subscriptions** into the top search bar and open it. You must see at least one subscription with the status Active. Note its name - you will select it in the next steps.

SHOT 1.2

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The Subscriptions blade listing the subscription name, ID and Active status. Blur or crop the subscription ID before sharing.

Figure 1.2 - Subscriptions blade.

What to check before moving on

Status must read Active. If it reads Disabled or Expired, stop here and resolve it - nothing in this lab will work without an active subscription.

Privacy note for the instructor

Subscription IDs, tenant IDs and object IDs are identifiers, not secrets, but they should still be cropped or blurred before a screenshot is shared with a class or uploaded anywhere. Make this a visible habit - students copy what they see you do.

3 Create a resource group

Search for **Resource groups** and select **Create**. Fill in the subscription, the resource group name from the naming table, and the region. Then move to **Review + create** and confirm.

SHOT 1.3

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The Create a resource group form filled in, showing subscription, resource group name and region fields.

Figure 1.3 - Creating a resource group.

What to check before moving on

The region you pick here is only metadata for the group itself - it does not restrict where resources inside it can live. Students ask this every single time, so address it before they do.

Why the region field is confusing

A resource group has a region because Azure stores the group's own metadata somewhere. But you can place a virtual machine from Central India and a storage account from West Europe inside the same resource group. The group's region does not constrain its contents. It only matters in one situation: if that region has a complete outage, you may be unable to modify the group's metadata, even though the resources inside keep running.

4 Start creating the virtual machine

Search for **Virtual machines** and select **Create**, then **Azure virtual machine**. On the Basics tab, set the subscription, select the resource group you just created, and enter the virtual machine name and region from the naming table.

SHOT 1.4

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The Basics tab of the Create a virtual machine wizard with subscription, resource group, VM name and region filled in.

Figure 1.4 - Create a virtual machine, Basics tab (upper half).

What to check before moving on

The resource group dropdown must show the group from Step 3. If it is empty, the group was created under a different subscription.

5 Choose the image, size and authentication

Scroll down on the same Basics tab. Select the Ubuntu image and the B1s size. Under Administrator account choose **SSH public key**, set the username, and let Azure generate a new key pair.

SHOT 1.5

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Lower half of the Basics tab showing image, size, authentication type, username and the inbound port rules section.

Figure 1.5 - Image, size and authentication settings.

What to check before moving on

Confirm the size shows B1s and that the estimated monthly cost is displayed next to it. Point this cost out to the class - it is the first time most of them see that infrastructure has a visible price tag.

Teaching point: password versus SSH key

Azure offers both. SSH key authentication is the correct choice and the one the exam expects for Linux. A password-authenticated server exposed to the internet on port 22 will begin receiving automated login attempts within hours. This is worth stating explicitly rather than leaving students to discover it.

6 Review and create

Move through to **Review + create**. Azure runs a validation pass. Read the summary properly rather than clicking through it - the estimated cost and the full resource list are both shown here.

SHOT 1.6

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The Review + create tab showing Validation passed, the estimated hourly price, and the configuration summary.

Figure 1.6 - Validation and cost summary before deployment.

What to check before moving on

It must say Validation passed. If validation fails, the error names the exact field that is wrong - read it rather than starting over.

7 Download the private key

Because you chose to generate a new key pair, Azure will prompt you to download the private key file. This is offered exactly once. Download it and keep it safe - without it you cannot connect to the virtual machine.

SHOT 1.7

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The Generate new key pair dialog with the Download private key and create resource button.

Figure 1.7 - The one and only chance to download the private key.

What to check before moving on

If a student misses this dialog, the key is gone. The VM will still deploy, but they cannot connect to it. Warn the class before this step, not after.

8 Watch the deployment

Deployment usually takes one to three minutes. Open the deployment details and expand the operations list. This is the most instructive screen in the whole lab - you can watch Azure create the network, the interface, the public IP, the security group and the disk one by one.

SHOT 1.8

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The deployment progress page with the Operation details tab expanded, showing each sub-resource and its status.

Figure 1.8 - One request, seven resources.

What to check before moving on

Ask the class to count the resources being created. Most will have assumed a virtual machine is a single object. This screen corrects that assumption better than any slide.

9 Open the virtual machine and read its overview

Select **Go to resource**. The Overview blade is where you will spend most of your time with a virtual machine. Locate the status, the public IP address, the private IP address, the size and the resource group.

SHOT 1.9

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The virtual machine Overview blade showing Running status, public IP, private IP, size and location.

Figure 1.9 - Virtual machine Overview blade.

What to check before moving on

Status must read Running. Note both IP addresses - the public one is how you reach it from outside, the private one is how other resources in the same virtual network reach it. The difference between these two matters in the networking module.

Connect to the virtual machine

From your own terminal, using the key file you downloaded. Replace the placeholder with the public IP from the Overview blade.

Terminal - SSH connection

```
chmod 400 vm-lab01_key.pem
ssh -i vm-lab01_key.pem azureuser@<public-ip-address>

# Once connected, confirm where you are:
hostnamectl
df -h
```

If the connection hangs

The usual causes, in order of frequency: the network security group is not allowing port 22 inbound; the college network blocks outbound SSH; the key file permissions are too open; or the wrong username was used. Check them in that order. If the campus network blocks port 22 entirely, use the portal's built-in browser-based connection instead, which works over HTTPS.

Part B - The Same Task in Azure CLI

Everything you just did through roughly forty clicks can be done in three commands. This is not a party trick - it is how infrastructure is actually managed in production, because commands can be saved, reviewed, repeated and put into version control. Clicks cannot.

10 Open Cloud Shell

Select the Cloud Shell icon in the top toolbar of the portal. Choose **Bash** when prompted. Cloud Shell runs in the browser with the Azure CLI already installed and already authenticated - there is nothing to set up locally.

SHOT 1.10

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Cloud Shell open at the bottom of the portal with the Bash prompt visible.

Figure 1.10 - Cloud Shell in Bash mode.

What to check before moving on

On first use, Cloud Shell asks to create a storage account to persist your files. This is a small ongoing cost. For classroom use, the ephemeral session option is sufficient.

Verify your context first

Before creating anything, always confirm which subscription you are pointed at. This single habit prevents a large proportion of real-world mistakes.

Azure CLI

```
az account show --output table
az account list --output table

# If you need to switch:
az account set --subscription "<subscription-name>"
```

Create everything

Azure CLI

```
# 1. Create the resource group
az group create \
  --name rg-lab01-cli \
  --location centralindia

# 2. Create the virtual machine (Azure builds the network automatically)
az vm create \
  --resource-group rg-lab01-cli \
  --name vm-lab01-cli \
  --image Ubuntu2204 \
  --size Standard_B1s \
  --admin-username azureuser \
  --generate-ssh-keys \
  --output table

# 3. Open port 22 for SSH
az vm open-port \
  --resource-group rg-lab01-cli \
  --name vm-lab01-cli \
  --port 22
```

11 Compare the two results

Open the resource group list in the portal. You should now see two resource groups containing equivalent sets of resources - one built by clicking, one built by three commands.

SHOT 1.11

PASTE SCREENSHOT HERE

Resource groups list showing both the portal-created and CLI-created groups side by side.

Figure 1.11 - Same outcome, two methods.

What to check before moving on

Ask the class which method they would prefer if they had to build this fifty times. That question is the entire argument for infrastructure as code, and they will arrive at it themselves.

Verification

A lab is not complete because the screen looked right. Verify it.

Check	How to verify	Expected result
The resource group exists	<code>az group list --output table</code>	Both lab groups are listed
The VM is running	<code>az vm list -d --output table</code>	PowerState shows VM running
A public IP is assigned	<code>az vm list-ip-addresses --output table</code>	A public IP is shown
SSH works	Connect from your terminal	You reach a shell prompt on the VM
Supporting resources exist	<code>az resource list --resource-group rg-lab01-cli --output table</code>	Disk, NIC, public IP, NSG and VNet all listed

Common Errors and What They Actually Mean

Error or symptom	Cause	Fix
Quota exceeded for this region	The subscription's core limit for that VM family in that region is reached.	Choose a different region, choose a smaller size, or request a quota increase.
The requested size is not available in this region	Not every size exists in every region or zone.	Pick another size, or deploy to a different region.
Permission denied (publickey) on SSH	Wrong key file, wrong username, or file permissions too open.	Use <code>chmod 400</code> on the key, confirm the username is <code>azureuser</code> , confirm the key matches the VM.
Connection times out on port 22	The network security group is blocking it, or the campus network blocks outbound SSH.	Run <code>az vm open-port</code> , or use the portal's browser-based connection over HTTPS.
Subscription not found in Cloud Shell	Signed in to a directory that does not contain the subscription.	Switch directory in the portal, then reopen Cloud Shell.
The resource group name already exists	Names must be unique within a subscription.	Append your roll number as the naming table instructs.

Cleanup - Do Not Skip This

A B1s virtual machine left running for a month will consume a meaningful share of your free credit, and the disk continues to cost money even when the machine is stopped. Deleting the resource group removes every resource inside it in one action.

Azure CLI

```
az group delete --name rg-lab01-cli --yes --no-wait
az group delete --name rg-lab01-<your-roll-number> --yes --no-wait

# Confirm nothing is left:
az group list --output table
```

12 Confirm the deletion in the portal

Refresh the resource groups list. Both groups should be gone or in a Deleting state. Then open Cost Management and look at today's spend.

SHOT 1.12

PASTE SCREENSHOT HERE

Resource groups list after deletion, and the Cost Management view showing the small charge for this lab.

Figure 1.12 - Verified cleanup and the cost of the exercise.

What to check before moving on

Showing the actual cost of the lab - usually a very small amount - makes the billing model concrete in a way that no explanation does.

Stopped is not the same as deleted

Stopping a virtual machine from inside the operating system still bills you for compute. Deallocating it from the portal or CLI stops compute charges but the disk still costs money. Only deleting removes all charges. Students confuse these three states constantly - state the difference explicitly.

Lab Assessment

Submit the following. This is what will be marked.

- A screenshot of the virtual machine Overview blade showing Running status, with the subscription ID cropped out.
- A screenshot of the deployment operations list showing every resource that was created.

- The output of `az vm list -d --output table`.
- A screenshot confirming both resource groups were deleted.
- In two or three sentences: name every resource Azure created automatically, and explain why each one was necessary.

Questions to Answer Before the Next Lab

1. What is the difference between stopping, deallocating and deleting a virtual machine?
2. Why does a virtual machine need a network interface rather than connecting to the network directly?
3. What would happen to the virtual machine if you deleted only its resource group?
4. Why does the resource group have a region if its contents can be in other regions?
5. Name two advantages of the CLI method over the portal method.
6. Which of the seven created resources would continue to cost money if the VM were deallocated?

Next lab

Lab 02 - Azure Storage Accounts: creating a storage account, uploading to Blob storage, understanding the redundancy options, and generating a shared access signature.